Haoyang Jiang

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Education

- **Ph.D. in Data Science** *Expected Graduation: Dec. 2027*
  College of William & Mary, Williamsburg, VA
- **Ph.D. in Mechanical Engineering (Transferred)** *Aug. 2023 – Aug. 2024*
  University of Utah, Salt Lake City, UT
- **M.S. in Astrometry and Celestial Mechanics** *Sept. 2020 – Jun. 2023*
  University of Science and Technology of China, Hefei, China
- **B.S. in Geophysics (Minor in Computer Science)** *Sept. 2016 – Jun. 2020*
  China University of Mining and Technology, Beijing, China

Research Projects

- **Foundation Models for Forecasting and Data Assimilation**: Extending a foundation model with in-context assimilation for observation-guided long-horizon forecasting under sparse data and distribution shift.
- **Multi-Back Flow for Full-Waveform Inversion** *[GitHub]*: Establishing a clean separation between prior training and inverse-task inference, enabling generalization to new acquisition geometries and variable source counts without paired task-specific retraining.
- **Second-Order Optimal-Transport Flow Matching** *[GitHub]*: Deriving a second-order phase-space framework from a minimum-acceleration variational principle that unifies transport coupling, trajectories, terminal-velocity closure, and acceleration supervision, delivering gains across images, videos, and scientific systems.
- **Boundary-Aware GNNs with Ghost Nodes** *[GitHub]*: Introducing learned ghost-node proxies and physics-based refiners to recover missing external forcing, reduce boundary errors, and stabilize long-horizon forecasting on open physical networks.
- **Physics-Guided Neural Flux Prediction** *[GitHub]*: Proposing directional difference operators and physics-consistent message passing that preserve high-frequency flux information and distinguish forward from reversed topology on directed physical networks.

Publications

1. Jiang H. Y., Qu Y. Z. *Fredholm Integral Equations Neural Operator (FIE-NO) for Boundary Value Problems.* Accepted by **Machine Learning: Engineering**, 2026.
2. Jiang H. Y., Wang J. D., He Y., et al. *Topology-aware Neural Flux Prediction Guided by Physics*. In **ICML**, 2025.
3. Jiang H. Y., Zhang M. J., et al. *Orbital error propagation considering atmospheric density uncertainty.* **Advances in Space Research**, 71(6), 2566–2574, 2023.
4. Jiang H. Y., Qu Y. Z. *Transfer Operator Learning with Fusion Frame.* arXiv:2408.10458, 2024.

Honors and Awards

- **Outstanding Graduate Student Award**, University of Science and Technology of China, 2022, 2023
- **First-Class Scholarship**, China University of Mining and Technology (Beijing), 2018, 2019

Conferences for Invited Talks

- *Topology-Aware Graph Neural Networks by Physics.*
  KGML Workshop: Leading the New Paradigm of AI for Science, University of Michigan, Ann Arbor, Aug. 2025.
- *Maneuver detection in azimuth-elevation data of geosynchronous satellites using machine learning methods.*
  International Symposium on Space Flight Dynamics (ISSFD), Beijing, Aug. 2022.
- *Orbit error propagation considering atmospheric density uncertainty.*
  Youth Innovation Promotion Association of the Chinese Academy of Sciences, Nanjing, Dec. 2021.

Professional Service

- **Reviewer**: MM (2025), ICDM (2025, 2026), AAAI (2026), ICLR (2026), KDD (2026), NeurIPS (2026)

Internship Experience

ORIGIN SPACE, Nanjing, China *Feb. 2022 – Jun. 2022*

Assistant Engineer Intern

- Reproduced and benchmarked machine-learning and numerical methods on real-world engineering datasets.
- Improved research prototypes through data preprocessing, parameter tuning, and runtime optimization.

Skills

- **Machine Learning**: Neural Operators, Physics-Informed ML, Graph Neural Networks, Generative Models
- **Scientific Computing**: Numerical Algorithms, PDE/ODE Solvers, Inverse Problems
- **Programming**: Python, MATLAB, and relevant ML libraries